

8.4. Single_crystal: The single crystal component

Input Parameters for component Single_crystal from samples

1 <Parameter = value>, [Unit], Description

The **Single_crystal** component models a thick, flat single crystal with multiple scattering and absorption with elastic coherent scattering. An elastic incoherent background may also be simulated. It may be used to describe samples for diffraction, but also for accurate monochromator descriptions. The component is currently under further review. The current documentation is outdated, especially with respect to the model of crystal mosaicity.

The input parameters for the component are *xwidth*, *yheight*, and *zdepth* to define the dimensions of the crystal in meters (area is centered); *delta_d_d* to give the value of $\Delta d/d$ (no unit); (*ax*, *ay*, *az*), (*bx*, *by*, *bz*), and (*cx*, *cy*, *cz*) to define the axes of the direct lattice of the crystal (the sides of the unit cell) in units of Ångström; and *reflections*, a string giving the name of the file with the list of structure factors to consider. The mosaic is specified *either* isotropically as *mosaic*, *or* anisotropically as *mosaic_h* (rotation around the *Y* axis), *mosaic_v* (rotation around the *Z* axis), and *mosaic_n* (rotation around the *X* axis); in all cases in units of full-width-half-maximum minutes of arc.

Optionally, the absorption cross-section at 2200 m/s and the incoherent cross-section may be given as *absorption* and *incoherent* (in barns), with default of zero; and *p.transmit* may be assigned a fixed Monte Carlo probability for transmission through the crystal without any interaction.

The user must specify a list of reciprocal lattice vectors $\boldsymbol{\tau}$ to consider along with their structure factors $|F_{\boldsymbol{\tau}}|^2$. The user must also specify the coordinates (in direct space) of the unit cell axes $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$, from which the reciprocal lattice will be computed. See section 8.4.5 for file format specifications.

In addition to coherent scattering, **Single_crystal** also handles incoherent scattering and absorption. The incoherent scattering cross-section is supplied by the user as a constant σ_{inc} . The absorption cross-section is supplied by the user at 2200 m/s, so the actual cross-section for a neutron of velocity v is $\sigma_{\text{abs}} = \sigma_{2200} \frac{2200 \text{ m/s}}{v}$.

8.4.1. The physical model

The textbook expression for the scattering cross-section of a crystal is [Squ78, ch.3]:

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{coh.el.}} = N \frac{(2\pi)^3}{V_0} \sum_{\boldsymbol{\tau}} \delta(\boldsymbol{\tau} - \boldsymbol{\kappa}) |F_{\boldsymbol{\tau}}|^2 \quad (8.18)$$

Here $|F_{\boldsymbol{\tau}}|^2$ is the structure factor (defined in section 8.3), N is the number of unit cells, V_0 is the volume of an individual unit cell, and $\boldsymbol{\kappa} (= \mathbf{k}_i - \mathbf{k}_f)$ is the scattering vector. $\delta(\mathbf{x})$ is a 3-dimensional delta function in reciprocal space, so for given incoming wave vector $\mathbf{k}_i$ and lattice vector $\boldsymbol{\tau}$, only a single final wave vector $\mathbf{k}_f$ is allowed. In general, this wavevector will not fulfill the conditions for elastic scattering ($k_f = k_i$). In a real

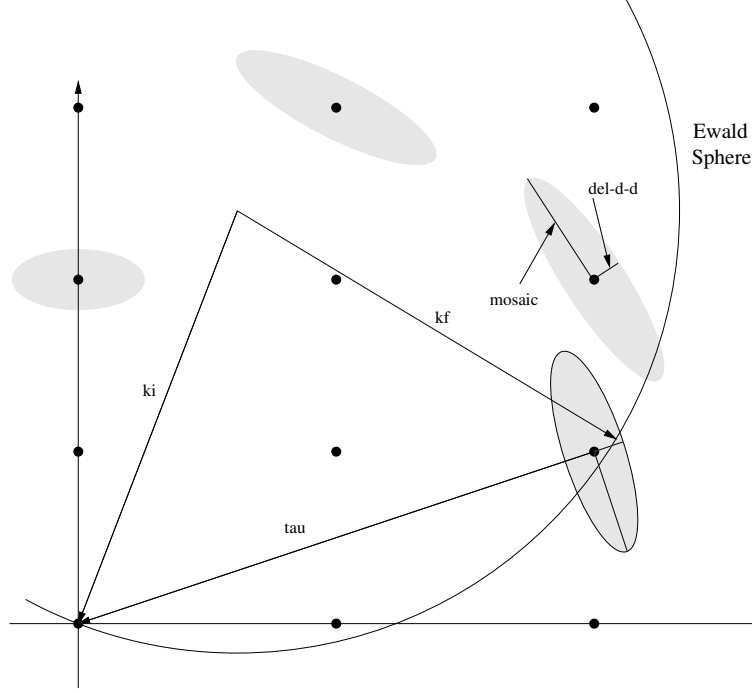


Figure 8.4.: Ewald sphere construction for a single neutron showing the Gaussian broadening of reciprocal lattice points in their local coordinate system.

crystal, however, reflections are not perfectly sharp. Because of imperfection and finite-size effects, there will be a small region around τ in reciprocal space of possible scattering vectors.

Single_crystal simulates a crystal with a mosaic spread η and a lattice plane spacing uncertainty $\Delta d/d$. In such crystals the reflections will not be completely sharp; there will be a small region around each reciprocal lattice point of the crystal that contains valid scattering vectors.

We model the mosaicity and $\Delta d/d$ of the crystal with 3-dimensional Gaussian functions in reciprocal space (see figure 8.4). Two of the axes of the Gaussian are perpendicular to the reciprocal lattice vector τ and model the mosaicity. The third one is parallel to τ and models $\Delta d/d$. We assume that the mosaicity is small so that the possible directions of the scattering vector may be approximated with a Gaussian in rectangular coordinates.

If the mosaic is isotropic (the same in all directions), the two Gaussian axes perpendicular to τ are simply arbitrary normal vectors of equal length given by the mosaic. But if the mosaic is anisotropic, the two perpendicular axes will in general be different for each scattering vector. In the absence of anything better, **Single_crystal** uses a model which is at least mathematically plausible and which works as expected in the two common cases: (1) isotropic mosaic, and (2) two mosaic directions (“horizontal and

vertical mosaic”) perpendicular to a scattering vector.

The basis for the model is a three-dimensional Gaussian distribution in Euler angles giving the orientation probability distribution for the micro-crystals; that is, the mis-orientation is given by small rotations around the X , Y , and Z axes, with the rotation angles having (in general different) Gaussian probability distributions. For given scattering vector $\boldsymbol{\tau}$, a rotation of the micro-crystals around an axis parallel to $\boldsymbol{\tau}$ has no effect on the direction of the scattering vector. Suppose we form the intersection between the three-dimensional Gaussian in Euler angles and a plane through the origin perpendicular to $\boldsymbol{\tau}$. This gives a two-dimensional Gaussian, say with axes defined by unit vectors $\mathbf{g}_1$ and $\mathbf{g}_2$ and mosaic widths η_1 and η_2 .

We now let the mosaic for $\boldsymbol{\tau}$ be defined by rotations around $\mathbf{g}_1$ and $\mathbf{g}_2$ with angles having Gaussian distributions of widths η_1 and η_2 . Since $\mathbf{g}_1$, $\mathbf{g}_2$, and $\boldsymbol{\tau}$ are perpendicular, a small rotation of $\boldsymbol{\tau}$ around $\mathbf{g}_1$ will change $\boldsymbol{\tau}$ in the direction of $\mathbf{g}_2$. The two axes of the Gaussian mosaic in reciprocal space that are perpendicular to $\boldsymbol{\tau}$ will thus be given by $\tau\eta_2\mathbf{g}_1$ and $\tau\eta_1\mathbf{g}_2$.

We now derive a quantitative expression for the scattering cross-section of the crystal in the model. For this, we introduce a *local coordinate system* for each reciprocal lattice point $\boldsymbol{\tau}$ and use $\mathbf{x}$ for vectors written in local coordinates. The origin is $\boldsymbol{\tau}$, the first axis is parallel to $\boldsymbol{\tau}$ and the other two axes are perpendicular to $\boldsymbol{\tau}$. In the local coordinate system, the 3-dimensional Gaussian is given by

$$G(x_1, x_2, x_3) = \frac{1}{(\sqrt{2\pi})^3} \frac{1}{\sigma_1\sigma_2\sigma_3} e^{-\frac{1}{2}\left(\frac{x_1^2}{\sigma_1^2} + \frac{x_2^2}{\sigma_2^2} + \frac{x_3^2}{\sigma_3^2}\right)} \quad (8.19)$$

The axes of the Gaussian are $\sigma_1 = \tau\Delta d/d$ and $\sigma_2 = \sigma_3 = \eta\tau$. Here we used the assumption that η is small, so that $\tan \eta \approx \eta$ (with η given in radians). By introducing the diagonal matrix

$$D = \begin{pmatrix} \frac{1}{2}\sigma_1^2 & 0 & 0 \\ 0 & \frac{1}{2}\sigma_2^2 & 0 \\ 0 & 0 & \frac{1}{2}\sigma_3^2 \end{pmatrix}$$

equation (8.19) can be written as

$$G(\mathbf{x}) = \frac{1}{(\sqrt{2\pi})^3} \frac{1}{\sigma_1\sigma_2\sigma_3} e^{-\mathbf{x}^T D \mathbf{x}} \quad (8.20)$$

again with $\mathbf{x} = (x_1, x_2, x_3)$ written in local coordinates.

To get an expression in the coordinates of the reciprocal lattice of the crystal, we introduce a matrix U such that if $\mathbf{y} = (y_1, y_2, y_3)$ are the global coordinates of a point in the crystal reciprocal lattice, then $U(\mathbf{y} + \boldsymbol{\tau})$ are the coordinates in the local coordinate system for $\boldsymbol{\tau}$. The matrix U is given by

$$U^T = (\hat{u}_1, \hat{u}_2, \hat{u}_3),$$

where $\hat{u}_1$, $\hat{u}_2$, and $\hat{u}_3$ are the axes of the local coordinate system, written in the global coordinates of the reciprocal lattice. Thus $\hat{u}_1 = \boldsymbol{\tau}/\tau$, and $\hat{u}_2$ and $\hat{u}_3$ are unit vectors

perpendicular to $\hat{u}_1$ and to each other. The matrix U is unitarian, that is $U^{-1} = U^T$. The translation between global and local coordinates is

$$\mathbf{x} = U(\mathbf{y} + \boldsymbol{\tau}) \quad \mathbf{y} = U^T \mathbf{x} - \boldsymbol{\tau}$$

The expression for the 3-dimensional Gaussian in global coordinates is

$$G(\mathbf{y}) = \frac{1}{(\sqrt{2\pi})^3} \frac{1}{\sigma_1 \sigma_2 \sigma_3} e^{-(U(\mathbf{y}+\boldsymbol{\tau}))^T D (U(\mathbf{y}+\boldsymbol{\tau}))} \quad (8.21)$$

The elastic coherent cross-section is then given by

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{coh.el.}} = N \frac{(2\pi)^3}{V_0} \sum_{\boldsymbol{\tau}} G(\boldsymbol{\tau} - \boldsymbol{\kappa}) |F_{\boldsymbol{\tau}}|^2 \quad (8.22)$$

8.4.2. The algorithm

The overview of the algorithm used in the Single_crystal component is as follows:

1. Check if the neutron intersects the crystal. If not, no action is taken.
2. Search through a list of reciprocal lattice points of interest, selecting those that are close enough to the Ewald sphere to have a non-vanishing scattering probability. From these, compute the total coherent cross-section σ_{coh} (see below), the absorption cross-section $\sigma_{\text{abs}} = \sigma_{2200} \frac{2200 \text{ m/s}}{v}$, and the total cross-section $\sigma_{\text{tot}} = \sigma_{\text{coh}} + \sigma_{\text{inc}} + \sigma_{\text{abs}}$.
3. The transmission probability is $\exp(-\frac{\sigma_{\text{tot}}}{V_0} \ell)$ where ℓ is the length of the flight path through the crystal. A Monte Carlo choice is performed to determine whether the neutron is transmitted. Optionally, the user may set a fixed Monte Carlo probability for the first scattering event, for example to boost the statistics for a weak reflection.
4. For non-transmission, the position at which the neutron will interact is selected from an exponential distribution. A Monte Carlo choice is made of whether to scatter coherently or incoherently. Absorption is treated by weight adjustment (see below).
5. For incoherent scattering, the outgoing wave vector $\mathbf{k}_f$ is selected with a random direction.
6. For coherent scattering, a reciprocal lattice vector is selected by a Monte Carlo choice, and $\mathbf{k}_f$ is found (see below).
7. Adjust the neutron weight as dictated by the Monte Carlo choices made.
8. Repeat from [\[2\]](#) until the neutron is transmitted (to simulate multiple scattering).

For point [\[2\]](#), the distance *dist* between a reciprocal lattice point and the Ewald sphere is considered small enough to allow scattering if it is less than five times the maximum axis of the Gaussian, $\text{dist} \leq 5 \max(\sigma_1, \sigma_2, \sigma_3)$.

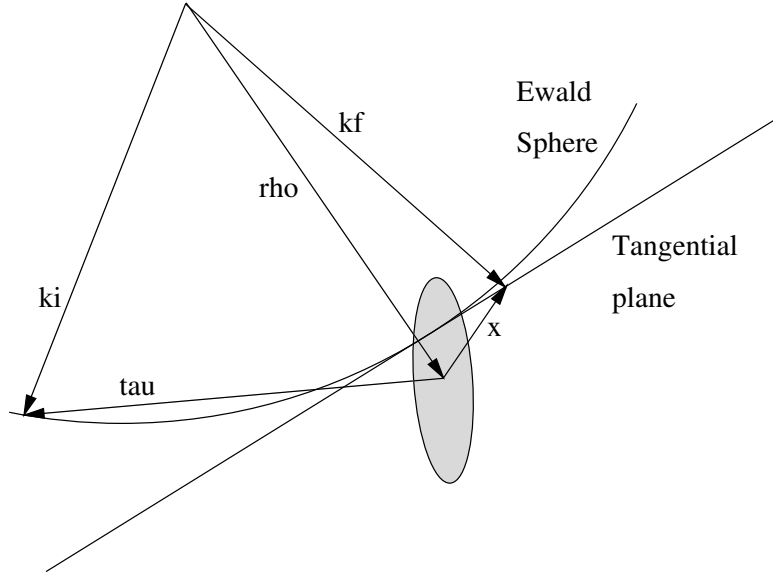


Figure 8.5.: The scattering triangle in the single crystal.

8.4.3. Choosing the outgoing wave vector

The final wave vector $\mathbf{k}_f$ must lie on the intersection between the Ewald sphere and the Gaussian ellipsoid. Since η and $\Delta d/d$ are assumed small, the intersection can be approximated with a plane tangential to the sphere, see figure 8.5. The tangential point is taken to lie on the line between the center of the Ewald sphere $-\mathbf{k}_i$ and the reciprocal lattice point $\boldsymbol{\tau}$. Since the radius of the Ewald sphere is k_i , this point is

$$\mathbf{o} = (k_i/\rho - 1)\boldsymbol{\rho} - \boldsymbol{\tau}$$

where $\boldsymbol{\rho} = \mathbf{k}_i - \boldsymbol{\tau}$.

The equation for the plane is

$$\mathbf{P}(\mathbf{t}) = \mathbf{o} + B\mathbf{t}, \quad \mathbf{t} \in \mathbb{R}^2 \quad (8.23)$$

Here $B = (\mathbf{b}_1, \mathbf{b}_2)$ is a 3×2 matrix with the two generators for the plane $\mathbf{b}_1$ and $\mathbf{b}_2$. These are (arbitrary) unit vectors in the plane, being perpendicular to each other and to the plane normal $\mathbf{n} = \boldsymbol{\rho}/\rho$.

Each $\mathbf{t}$ defines a potential final wave vector $\mathbf{k}_f(\mathbf{t}) = \mathbf{k}_i + \mathbf{P}(\mathbf{t})$. The value of the 3-dimensional Gaussian for this $\mathbf{k}_f$ is

$$G(\mathbf{x}(\mathbf{t})) = \frac{1}{(\sqrt{2\pi})^3} \frac{1}{\sigma_1 \sigma_2 \sigma_3} e^{-\mathbf{x}(\mathbf{t})^T D \mathbf{x}(\mathbf{t})} \quad (8.24)$$

where $\mathbf{x}(\mathbf{t}) = \boldsymbol{\tau} - (\mathbf{k}_i - \mathbf{k}_f(\mathbf{t}))$ is given in local coordinates for $\boldsymbol{\tau}$. It can be shown that equation (8.24) can be re-written as

$$G(\mathbf{x}(\mathbf{t})) = \frac{1}{(\sqrt{2\pi})^3} \frac{1}{\sigma_1 \sigma_2 \sigma_3} e^{-\alpha} e^{-(\mathbf{t}-\mathbf{t}_0)^T M (\mathbf{t}-\mathbf{t}_0)} \quad (8.25)$$

where $M = B^T DB$ is a 2×2 symmetric and positive definite matrix, $\mathbf{t}_0 = -M^{-1}B^T D\mathbf{o}$ is a 2-vector, and $\alpha = -\mathbf{t}_0^T M \mathbf{t}_0 + \mathbf{o}^T D\mathbf{o}$ is a real number. Note that this is a two-dimensional Gaussian (not necessarily normalized) in $\mathbf{t}$ with center $\mathbf{t}_0$ and axis defined by M .

To choose $\mathbf{k}_f$ we sample $\mathbf{t}$ from the 2-dimensional Gaussian distribution (8.25). To do this, we first construct the Cholesky decomposition of the matrix $(\frac{1}{2}M^{-1})$. This gives a 2×2 matrix L such that $LL^T = \frac{1}{2}M^{-1}$ and is possible since M is symmetric and positive definite. It is given by

$$L = \begin{pmatrix} \sqrt{\nu_{11}} & 0 \\ \frac{\nu_{12}}{\sqrt{\nu_{11}}} & \sqrt{\nu_{22} - \frac{\nu_{12}^2}{\nu_{11}}} \end{pmatrix} \quad \text{where } \frac{1}{2}M^{-1} = \begin{pmatrix} \nu_{11} & \nu_{12} \\ \nu_{12} & \nu_{22} \end{pmatrix}$$

Now let $\mathbf{g} = (g_1, g_2)$ be two random numbers drawn from a Gaussian distribution with mean 0 and standard deviation 1, and let $\mathbf{t} = L\mathbf{g} + \mathbf{t}_0$. The probability of a particular $\mathbf{t}$ is then

$$P(\mathbf{t})d\mathbf{t} = \frac{1}{2\pi} e^{-\frac{1}{2}\mathbf{g}^T \mathbf{g}} d\mathbf{g} \quad (8.26)$$

$$= \frac{1}{2\pi \det L} e^{-\frac{1}{2}(L^{-1}(\mathbf{t}-\mathbf{t}_0))^T (L^{-1}(\mathbf{t}-\mathbf{t}_0))} d\mathbf{t} \quad (8.27)$$

$$= \frac{1}{2\pi \det L} e^{-(\mathbf{t}-\mathbf{t}_0)^T M (\mathbf{t}-\mathbf{t}_0)} d\mathbf{t} \quad (8.28)$$

where we used that $\mathbf{g} = L^{-1}(\mathbf{t} - \mathbf{t}_0)$ so that $d\mathbf{g} = \frac{1}{\det L} d\mathbf{t}$. This is just the normalized form of (8.25). Finally we set $\mathbf{k}'_f = \mathbf{k}_i + \mathbf{P}(\mathbf{t})$ and $\mathbf{k}_f = (k_i/k'_f)\mathbf{k}'_f$ to normalize the length of $\mathbf{k}_f$ to correct for the (small) error introduced by approximating the Ewald sphere with a plane.

8.4.4. Computing the total coherent cross-section

To determine the total coherent scattering cross-section, the differential cross-section must be integrated over the Ewald sphere:

$$\sigma_{\text{coh}} = \int_{\text{Ewald}} \left(\frac{d\sigma}{d\Omega} \right)_{\text{coh.el.}} d\Omega$$

For small mosaic we may approximate the sphere with the tangential plane, and we thus get from (8.22) and (8.25):

$$\sigma_{\text{coh},\tau} = \int N \frac{(2\pi)^3}{V_0} G(\boldsymbol{\tau} - \boldsymbol{\kappa}) |F_{\boldsymbol{\tau}}|^2 d\Omega \quad (8.29)$$

$$= \frac{1}{\mathbf{k}_i^2} N \frac{(2\pi)^3}{V_0} \frac{1}{(\sqrt{2\pi})^3} \frac{e^{-\alpha}}{\sigma_1 \sigma_2 \sigma_3} |F_{\boldsymbol{\tau}}|^2 \int e^{-(\mathbf{t}-\mathbf{t}_0)^T M (\mathbf{t}-\mathbf{t}_0)} d\mathbf{t} \quad (8.30)$$

$$= \det(L) \frac{1}{\mathbf{k}_i^2} N \frac{(2\pi)^{3/2}}{V_0} \frac{e^{-\alpha}}{\sigma_1 \sigma_2 \sigma_3} |F_{\boldsymbol{\tau}}|^2 \int e^{-\frac{1}{2} \mathbf{g}^T \mathbf{g}} d\mathbf{g} \quad (8.31)$$

$$= 2\pi \det(L) \frac{1}{\mathbf{k}_i^2} N \frac{(2\pi)^{3/2}}{V_0} \frac{e^{-\alpha}}{\sigma_1 \sigma_2 \sigma_3} |F_{\boldsymbol{\tau}}|^2 \quad (8.32)$$

$$= \frac{\det(L)}{\mathbf{k}_i^2} N \frac{(2\pi)^{5/2}}{V_0} \frac{e^{-\alpha}}{\sigma_1 \sigma_2 \sigma_3} |F_{\boldsymbol{\tau}}|^2 \quad (8.33)$$

$$\sigma_{\text{coh}} = \sum_{\boldsymbol{\tau}} \sigma_{\text{coh},\tau} \quad (8.34)$$

As before, we let $\mathbf{g} = L^{-1}(\mathbf{t} - \mathbf{t}_0)$ so that $d\mathbf{t} = \det(L) d\mathbf{g}$.

Neutron weight factor adjustment We now calculate the correct neutron weight adjustment for the Monte Carlo choices made. In three cases is a Monte Carlo choice made with a probability different from the probability of the corresponding physical event: When deciding whether to transmit the neutron or not, when simulating absorption, and when selecting the reciprocal lattice vector $\boldsymbol{\tau}$ to scatter from.

If the user has chosen a fixed transmission probability $f(\text{transmit}) = p_{\text{transmit}}$, the neutron weight must be adjusted by

$$\pi(\text{transmit}) = \frac{P(\text{transmit})}{f(\text{transmit})}$$

where $P(\text{transmit}) = \exp(-\frac{\sigma_{\text{tot}}}{V_0} \ell)$ is the physical transmission probability. Likewise, for non-transmission the adjustment is

$$\pi(\text{no transmission}) = \frac{1 - P(\text{transmit})}{1 - f(\text{transmit})}.$$

Absorption is never explicitly simulated, so the Monte Carlo probability of coherent or incoherent scattering is $f(\text{coh}) + f(\text{inc}) = 1$. The physical probability of coherent or incoherent scattering is

$$P(\text{coh}) + P(\text{inc}) = \frac{\sigma_{\text{coh}} + \sigma_{\text{inc}}}{\sigma_{\text{tot}}},$$

so again a weight adjustment $\pi(\text{coh}|\text{inc}) = \Pi(\text{coh}|\text{inc})/f(\text{coh}|\text{inc})$ is needed.

When choosing the reciprocal lattice vector $\boldsymbol{\tau}$ to scatter from, the relative probability for $\boldsymbol{\tau}$ is $r_{\boldsymbol{\tau}} = \sigma_{\text{coh},\boldsymbol{\tau}}/|F_{\boldsymbol{\tau}}|^2$. This is done to get better statistics for weak reflections. The Monte Carlo probability for the reciprocal lattice vector $\boldsymbol{\tau}$ is thus

$$f(\boldsymbol{\tau}) = \frac{r_{\boldsymbol{\tau}}}{\sum_{\boldsymbol{\tau}} r_{\boldsymbol{\tau}}}$$

whereas the physical probability is $P(\tau) = \sigma_{\text{coh},\tau}/\sigma_{\text{coh}}$. A weight adjustment is thus needed of

$$\pi(\tau) = \frac{P(\tau)}{f(\tau)} = \frac{\sigma_{\text{coh},\tau} \sum_{\tau} r_{\tau}}{\sigma_{\text{coh}} r_{\tau}}.$$

In most cases, however, only one reflection is possible, whence $\pi = 1$.

8.4.5. Implementation details

The equations describing **Single_crystal** are quite complex, and consequently the code is fairly sizeable. Most of it is just the expansion of the vector and matrix equations in individual coordinates, and should thus be straightforward to follow.

The implementation pre-computes a lot of the necessary values in the **INITIALIZE** section. It is thus actually very efficient despite the complexity. If the list of reciprocal lattice points is big, however, the search through the list will be slow. The precomputed data is stored in the structures **hkl_info** and in an array of **hkl_data** structures (one for each reciprocal lattice point in the list). In addition, for every neutron event an array of **tau_data** is computed with one element for each reciprocal lattice point close to the Ewald sphere. Except for the search for possible τ vectors, all computations are done in local coordinates using the matrix U to do the necessary transformations.

The list of reciprocal lattice points is specified in an ASCII data file. Each line contains seven numbers, separated by white space. The first three numbers are the (h, k, l) indices of the reciprocal lattice point, and the last number is the value of the structure factor $|F_{\tau}|^2$, in barns. The middle three numbers are not used and may be omitted; they are nevertheless recommended since this makes the file format compatible with the output from the Crystallographica program **Cry**. Any line beginning with any character of **#** or **%** is considered to be a comment, and lines which can not be read as vectors/matrices are ignored.

The column signification may also explicitly be set in the data file header using any of the lines:

```
1 #column_h <index of the Bragg Qh column>
2 #column_k <index of the Bragg Qk column>
3 #column_l <index of the Bragg Ql column>
4 #column_F2 <index of the squared str. factor '|F|^2' column [b]>
5 #column_F <index of the structure factor norm '|F|' column>
```

Other component parameters may as well be specified in the data file header with lines e.g.:

```
1 #sigma_abs <value of Absorption cross section [barns]>
2 #sigma_inc <value of Incoherent cross section [barns]>
3 #Delta_d/d <value of Delta_d/d width for all lines>
4 #lattice_a <value of the a lattice parameter [Angs]>
5 #lattice_b <value of the b lattice parameter [Angs]>
6 #lattice_c <value of the c lattice parameter [Angs]>
7 #lattice_aa <value of the alpha lattice angle [deg]>
8 #lattice_bb <value of the beta lattice angle [deg]>
9 #lattice_cc <value of the gamma lattice angle [deg]>
```


Example data *.lau files are given in directory MCSTAS/data.

These files contain an extensive self-documented header defining most the sample parameters, so that only the file name and mosaicity should be given to the component:

```
1 Single_crystal(xwidth=0.01, yheight=0.01, zdepth=0.01,  
2 mosaic = 5, reflections="YBaCuO.lau")
```

Powder files from ICSD/LAZY [Ics] and Fullprof [Ful] may also be used (see Table 1.2, page 15). We do not recommend to use these as the equivalent $\vec{q}$ vectors are superposed, not all Bragg spots will be simulated, and the intensity will not be scaled by the multiplicity for each spot.